

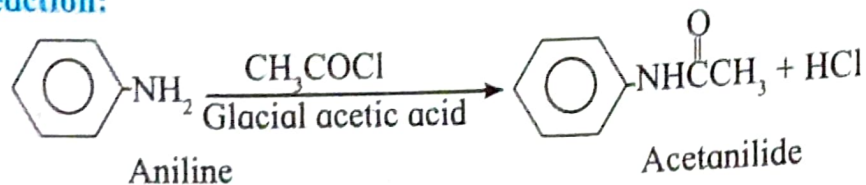
Experiment No. 16

Date: / /

Aim: To prepare acetanilide from acetyl chloride.

Theory: The replacement of one hydrogen atom of the $-NH_2$ group of aniline by CH_3-CO- group in the presence of glacial acetic acid gives acetanilide. Acetylation with acetyl chloride is carried out in the presence of glacial acetic acid.

Chemicals Reaction:



Apparatus: Boiling tube, water bath, beaker, funnel, etc.

Chemicals: Aniline, acetic acid, acetyl chloride.

Procedure:

1. Take 1 mL of aniline in a dry boiling tube, add 1 mL glacial acetic acid to it and mix the two solutions thoroughly.
2. To the above mixture, add 1 mL acetyl chloride in lots of 4-5 drops at a time with shaking. The mixture becomes warm. (If the boiling tube becomes unbearable to touch, cool it under tap water.)
3. After the complete addition of acetyl chloride, heat the mixture occasionally over the flame with shaking for about 15 minutes. Note that the reaction mixture should not boil.
4. Cool the boiling tube and add near about 10 mL of ice-cold water into the tube with constant stirring.
5. Filter the acetanilide obtained as white powder and wash with water till the filtrate is neutral to blue litmus paper.
6. Crystallize the crude acetanilide with hot water. White, shining, needle shaped crystals are obtained.
7. Dry the crystals, weigh it.

Observation and Result:

1. Yield of acetanilide = 3.14 g
2. Colour of acetanilide crystal = white
3. Shape of acetanilide crystals = needle shape

Note: Acetanilide can also be prepared by using acetic anhydride in presence of pyridine. But difficult to handle the corrosive chemicals as well as set up the apparatus/assembly.

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

1. The functional group present in acetanilide is
 a. oxime ☒ b. amide c. amino d. ketone
2. Preparation of acetanilide is known as
☒ a. acetylation b. ammonolysis
 c. alkylation d. peptide formation
3. The colour of acetanilide crystal is
☒ a. white b. blue c. yellow d. red
4. Shape of the crystals of acetanilide is
 a. triclinic b. monoclinic ☒ c. needle-shaped d. tetragonal
5. During preparation of acetanilide, replacement of hydrogen of aniline takes place by
 a. hydrogen atom ☒ b. acetyl group
 c. amino group d. carboxylic group

Short answer questions

1. Why acetyl chloride is added in a lot during preparation of acetanilide?

Ans.

The nucleophilic substitution of aromatic amines with acetyl chloride results in the formation of N-substituted amides.

2. Write an IUPAC name of acetanilide.

Ans.

N-phenyl acetanilide

3. Write the meaning of term acetylation?

Ans.

Acetylation is a chemical reaction in which a hydrogen atom is substituted by an acetyl group ($\text{CH}_3\text{C}(=\text{O})$).

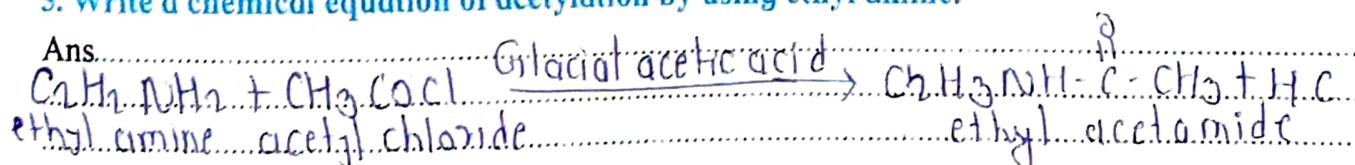
4. What is the role of glacial acetic acid in acylation reaction?

Ans.

Glacial acetic is used because it is an excellent polar protic solvent capable of dissolving acetanilide.

5. Write a chemical equation of acetylation by using ethyl amine.

Ans.



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